



UNIVERSITY OF ENGINEERING AND MANAGEMENT

(Established by Act XXV of 2014 of Govt. of West Bengal & recognized by UGC, Ministry of HRD, Govt. of India)
University Area, Plot No. 30-B/3, Main Arterial Road, New Town, Action Area - III, Kolkata - 700160, WB, India
Admission Office: 'ASHRAM', GN-34/2, Salt Lake Electronics Complex, Kolkata - 700091, WB, India

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Date: 24th April, 2025

It is hereby notified to all concerned persons of University of Engineering and Management Kolkata that the selection committee has approved the following Grant-In-Aid Projects of the faculty members on the basis of the overall performance in the interview held on 5th April, 2025 and the faculty members should submit the previous projects completion reports (even partially, if any) and novelty of the projects to the Accounts Department :

PROJECT GRANT No.	NAME	PROJECT NAME	APPROVED PROJECT GRANT BY THE EXPERT COMMITTEE MEMBERS (INR)
IEMT(N)/2025 /A/03 - G01	Prof. Apurba Nandi (PI), Prof. Sandip Mandal (Co-PI), Prof. Avik Kumar Das (Co-PI)	AI-Powered Drone Technology for Forest Fire Detection	Rs. 40,000/-
IEMT(N)/2025 /A/03 - G02	Prof. Apurba Nandi (PI), Prof. Avik Kumar Das (Co-PI), Prof. Sayanti Das (Co-PI), Dr. Sandip Mandal (Co-PI)	AI-Driven Riverbank Erosion Monitoring and Disaster Mitigation System	Rs. 30,000/-
IEMT(N)/2025 /A/03 - G03	Prof. Apurba Nandi (PI) Prof. Avik Kumar Das (Co-PI), Dr. Arjeet Ghosh (Co-PI), Dr. Sandip Mandal (Co-PI)	Bishnupur Historical Temples and Information Retrieval using Deep Learning Model and its hardware implementation	Rs. 30,000/-
IEMT(N)/2025 /A/03 - G04	Prof. Apurba Nandi (PI) Prof. Avik Kumar Das (Co-PI), Prof. Sandip Mandal (Co-PI)	Hasta Vidya : A contactless Gesture Control Interface	Rs. 40,000/-
IEMT(N)/2025 /A/03 - G05	Prof. Apurba Nandi (PI) Prof. Avik Kumar Das (Co-PI) Prof. Sandip Mandal (Co-PI)	GreenHack: IoT and Machine Learning based kit - a one-stop solution to contemporary and controlled farming	Rs. 40,000/-
IEMT(N)/2025 /A/03 - G06	Prof. Avik Kumar Das (PI) Dr. Moupiya Ghosh (Co-PI) Dr. Rina Dutta (Co-PI)	Integration of Artificial Intelligence with Nanotechnology	Rs. 1,50,000/-
IEMT(N)/2025 /A/03 - G07	Prof. Avik Kumar Das (PI) Prof. Suchanda Chatterjee (Das) (Co-PI), Prof. Dr. Sandip Mandal (Co-PI)	Machine Learning Model for Predicting Health Risks Using Vital Signs and Real Time Clinical Data	Rs. 50,000/-
IEMT(N)/2025 /A/03 - G08	Prof. Avik Kumar Das (PI) Prof. Apurba Nandi (PI) Dr. Sandip Mandal (Co-PI)	AI-Powered Underwater Sound Detection And Image Processing for sustainable underwater environment monitoring	Rs. 80,000/-
IEMT(N)/2025 /A/03 - G09	Prof. Avik Kumar Das (PI) Prof. Apurba Nandi (PI) Prof. Dr. Sandip Mandal (Co-PI)	Innovative Wheelchair Design for Enhanced Independence and Health Monitoring using AI & Hardware Integration	Rs. 60,000/-
IEMT(N)/2025 /A/03 - G10	Prof. Avik Kumar Das (PI) Prof. Apurba Nandi (PI) Prof. Dr. Sandip Mandal (Co-PI)	AI-Powered Underwater Monitoring and Exploration using Autonomous Underwater Vehicles (AUVs)	Rs. 60,000/-
IEMT(N)/2025 /A/03 - G11	Prof. Avik Kumar Das (PI) Prof. Dr. Arjeet Ghosh (Co-PI) Dr. Sandip Mandal (Co-PI)	Vayurakshak Disaster Management Drone Using AI And MI	Rs. 1,00,000/-
IEMT(N)/2025 /A/03 - G12	Prof. Avik Kumar Das (PI) Prof. Apurba Nandi (Co-PI) Dr. Sandip Mandal (Co-PI)	Wash out Vacuum based Flush System	Rs. 60,000/-

Other Institutes of the Group

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IEMT(N)/2025 /A/03 - G13	Prof. Avik Kumar Das (PI), Prof. Apurba Nandi (Co-PI), Dr. Sandip Mandal (Co-PI)	Gesture-Controlled Conveyor System Using BLDC Motor for Enhanced Automation and Safety	Rs. 50,000/-
IEMT(N)/2025 /A/03 - G14	Prof. Dr. Abhisek Halder (PI) Prof. Sankhadeep Chatterjee (Co-PI)	Investigation of Compressor oil-based Hybrid Nanolubricants for Enhanced Refrigeration Performance using Machine Learning techniques.	Rs. 50,000/-
IEMT(N)/2025 /A/03 - G15	Prof. Dr. Abir Chatterjee (PI) Prof. Nilanjan Chatterjee (Co- PI)	Temp Track RFID : Smart Temperature Monitoring Solution	Rs. 50,000/-
IEMT(N)/2025 /A/03 - G16	Prof. Dr. Abir Chatterjee (PI), Dr. Subhajit Das (Co-PI)	Hexa-Agro: AI Driven Hexa-copter for Precision Pest Management and Eco- Friendly Agriculture	Rs. 40,000/-
IEMT(N)/2025 /A/03 - G17	Prof. Dr. Anirban Ganguly (PI) Prof. Dr. Anirban Ganguly (Co- PI) Prof. (Dr.) Subhalaxmi Chakraborty (Co-PI), Prof. Debanjana Datta Mitra (Co-PI)	EdgeSentinel: AI-Enabled Patient Monitoring System with Edge Computing and Enhanced Security	Rs. 50,000/-
IEMT(N)/2025 /A/03 - G18	Prof. Dr. Anirban Ganguly (PI) Prof. Dr. Anirban Das (Co-PI)	IoT Solutions for Paralyzed Patient Care	Rs. 50,000/-
IEMT(N) /2025/A/03 - G19	Prof. Dr. Arijet Ghosh (PI) Prof. Apurba Nandi (Co-PI) Dr. Sandip Mandal (Co-PI)	"कृषक मित्र (Krishak Mitra)" - Transforming Agriculture with Edge Computing and Robotics	Rs. 1,50,000/-
IEMT(N) /2025/A/03 - G20	Prof. Dr. Arijet Ghosh Prof. Sangita Dutta (Co-PI) Dr. Sandip Mandal (Co-PI)	गरुड(Garudh): AI-Driven UAVs for Rapid Disaster Response and Long-Term Resilience Enhancement	Rs. 2,00,000/-
IEMT(N) /2025/A/03 - G21	Prof. Dr. Arijet Ghosh Prof. Avik Kumar Das (Co-PI) Dr. Sandip Mandal (Co-PI)	EcoDrone AI: AI-Powered Drones for Real- time Species Distribution Modeling and Ecosystem Monitoring	Rs. 1,00,000/-
IEMT(N) /2025/A/03 - G22	Prof. Dr. Arnab Ghosh (PI) Prof. Pradipta Sarkar (Co-PI)	Design of an Underwater Drone (ROVUS) for Surveillance and Monitoring of Marine Ecosystem to attain certain Sustainable Development Goals	Rs. 50,000/-
IEMT(N) /2025/A/03 - G23	Prof. Dr. Ashim Kumar Biswas (PI)	(Investigation and Design of a Non- conventional Agricultural Residue Rice Husk Based MIMO Antenna with Reduced Mutual Coupling)	Rs. 30,000/-
IEMT(N) /2025/A/03 - G24	Prof. Dr. Ashim Kumar Biswas (PI)	(Design and Investigation of a Highly Decoupled Biodegradable Jute-substrate Based 5G MIMO Antenna)	Rs. 19,000/-
IEMT(N)/2025 /A/03 - G25	Prof. Dr. Ashim Kumar Biswas (PI)	(Development of a Low-Cost Highly Decoupled Four-element Textile MIMO Antenna with Two Kinds of Asymmetric Radiators)	Rs. 27,000/-
IEMT(N) /2025/A/03 - G26	Prof. Dr. Abhijit Debnath (PI) Prof. Ranadhir Das (Co-PI) Prof. Bapi Biswas (Co-PI) Prof. (Dr.) Sudipta Dey (Co-PI)	Rural vegetation index prediction and subsequent climate mitigation with the application of Deep Learning	Rs. 50,000/-

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IEMT(N)/2025 /P/03 – G43	Prof. Prakash Banerjee (PI) Dr. Abir Chatterjee (co-PI)	Revolutionizing Geriatric Care: An AI-Enhanced Stroke Detection System with IoT-Based Alerts and Panic Button Integration	Rs. 13,500/-
IEMT(N)/2025 /P/03 – G44	Prof. Prakash Banerjee (PI) Dr. Abir Chatterjee (co-PI)	Flammable Gas and Fire Detection using IoT Based System for Process Industry Application.	Rs. 13,500/-
IEMT(N)/2025 /P/03 – G45	Prof. Dr. Priyanka Bera (PI) Prof. Pubali De (Co-PI) Prof. Sananda Pal (Co-PI)	AI driven wellness Insights: Harnessing Image-Based PPG for Advanced Health Information Extraction	Rs. 30,000/-
IEMT(N)/2025 /P/03 – G46	Prof. Pradipta Sarkar (PI) Dr. Srilekh Mukherjee (Co-PI)	Automated Sound-Based Object Searching Hardware Module	Rs. 50,000/-
IEMT(N)/2025 /P/03 – G47	Prof. Pradipta Sarkar (PI) Dr. Srilekh Mukherjee (Co-PI)	LANDEASE	Rs. 40,000/-
IEMT(N)/2025 /P/03 – G48	Prof. Pratik Talukder (PI)	Formulation of potential of hydroethanolic extracts of Asparagus racemosus as an effective physiologically viable strategy	Rs. 90,000/-
IEMT(N)/2025 /P/03 – G49	Prof. Dr. Rupsa Bhattacharya (PI) Dr. Dibyajit Lahiri (Co-PI), Dr. Moupiya Nag (Co-PI).	Exploration of bioactive compounds from Tinospora cordifolia for their antibiofilm and healing burn wound infection caused by two clinical isolates	Rs. 3,00,000/-
IEMT(N)/2025 /P/03 – G50	Prof. Rangon Sarkar (PI) Dr. Sandip Mandal (Co-PI), Dr. Susmita Biswas (Co-PI)	Comparison of RF and Analog parameters of Graphene Nanoribbon MOSFET and Conventional MOSFET using SILVACO ATLAS TCAD Tool	Rs. 40,000/-
IEMT(N)/2025 /S/03 – G51	Prof. Dr. Shreya Nag (PI) Prof. Dr. Abir Chatterjee (Co-PI)	Electronic Nose Based Universal Gas Detection System	Rs. 40,000/-
IEMT(N)/2025 /S/03 – G52	Prof. Dr. Soniya Kundu (PI) Dr. Suvadra Das (Co-PI), Dr. Moupiya Ghosh (Co-PI)	Development of biopolymer-based films with carbon nanodots for smart and active food packaging applications	Rs. 80,000/-
IEMT(N)/2025 /S/03 – G53	Prof. Dr. Soumik Poddar (PI)	Development of Biogenic Metal-Organic Framework (MOF) for Prevention of Neurodegenerative Diseases	Rs. 50,000/-
IEMT(N)/2025 /S/03 – G54	Prof. Dr. Susmita Mukherjee (PI) Prof. Sonali Pal (Co-PI)	Accumulation of Heavy metal in potato (Solanum indicum) grown in industrial area of West Bengal and the remediation through ion-exchange method of Layered Double Hydroxide	Rs. 1,00,000/-
IEMT(N)/2025 /S/03 – G55	Prof. Sangita Dutta (PI) Prof. Sweta Saha (Co-PI), Prof. Dr. Sandip Mandal (Co-PI)	EDGE-GUARD: Real-Time Deepfake Detection using Edge Computing	Rs. 30,000/-
IEMT(N)/2025 /S/03 – G56	Prof. Sangita Dutta (PI) Prof. Dr. Siddhartha Roy (Co-PI), Prof. Dr. Sandip Mandal (Co-PI)	PhishShield : Real-Time Email Phishing Detection Using NLP and Metadata Analysis	Rs. 1,00,000/-
IEMT(N)/2025 /S/03 – G57	Prof. Sangita Dutta (PI) Prof. Dr. Siddhartha Roy (Co-PI), Prof. Dr. Sandip Mandal (Co-PI)	FameBot: AI-Powered Conversations with Legendary Figures	Rs. 1,00,000/-

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IEMT(N)/2025 /S/03 - G58	Prof. Sangita Dutta (PI) Prof. Apurba Nandi (Co-PI), Prof. Dr. Sandip Mandal (Co-PI)	Next-Gen Anomaly Detection System: Revolutionizing Predictive Maintenance and Operational Efficiency	Rs. 40,000/-
IEMT(N)/2025 /S/03 - G59	Prof. Sangita Dutta (PI) Prof. Dr. Arijet Ghosh (Co-PI), Dr. Sandip Mandal (Co-PI)	Wiscan : Analyze and Mitigate wifi vulnerabilities	Rs. 20,000/-
IEMT(N)/2025 /S/03 - G60	Prof. Sayantani Das (PI) Prof. Nitu Saha (Co-PI), Dr. Sandip Mandal (Co-PI)	ChoreoSpeak: A Visual-Sensory Communication System for Enhanced Accessibility	Rs. 15,000/-
IEMT(N)/2025 /S/03 - G61	Prof. Sayantani Das (PI) Prof. Apurba Nandi (Co-PI), Prof. Arijet Ghosh (Co-PI), Dr. Sandip Mandal (Co-PI)	Sign Language Detection and Translator: Hardware Implementation Using Mediapipe	Rs. 20,000/-
IEMT(N)/2025 /S/03 - G62	Prof. Sayantani Das (PI) Prof. Nitu Saha (Co-PI)	SecureMe : Empowering Safe Digital Experiences	Rs. 20,000/-
IEMT(N)/2025 /S/03 - G63	Prof. Sayantani Das (PI) Prof. Dr. Sandip Mandal (Co-PI), Prof. Apurba Nandi (Co-PI), Prof. Nitu Saha (Co-PI)	Smart Wheelchair: A Comprehensive IoT, AI, and Health Monitoring Solution	Rs. 60,000/-
IEMT(N)/2025 /S/03 - G64	Prof. Sramana Mukherjee (PI)	Hydro Float : Drone-based Garbage Detection and Collection System for water bodies	Rs. 1,00,000/-
IEMT(N)/2025 /S/03 - G65	Prof. Subrata Mondal (PI)	Material characterization of developed Graphene Reinforced Aluminium Metal Matrix Composites for Advanced Tool Materials.	Rs. 40,000/-
IEMT(N)/2025 /S/03 - G66	Prof. Srilekha Mukherjee (PI) Prof. Pradipta Sarkar (Co-PI)	Solar guide Auto Parking	Rs. 40,000/-
IEMT(N)/2025 /S/03 - G67	Prof. Soumi Datta (PI) Prof. Sanchita De (Co-PI)	Sustainability through Green Entrepreneurship: Role of Ecotourism, Plastic Recycling, and Awareness Generation in West Bengal	Rs. 36,000/-
IEMT(N)/2025 /T/03 - G68	Prof. Dr. Tanay Pramanik	Development of green methodologies for the synthesis of Bis - Coumarin and their derivatives	Rs. 48,000/-
IEMT(N)/2025 /T/03 - G69	Prof. Dr. Tista Banerjee (PI) Prof. Gourab Saha (Co-PI) Dr. Souvik Maiti (Co-PI)	Development of an AI based intelligent bidirectional inverter-Converter System for Electric Vehicle	Rs. 70,000/-
Total			Rs. 44,27,000/-

Prof. (Dr.) Satyajit Chakrabarti
Pro-Vice-Chancellor

Copy to:

1. Chancellor
2. Vice-Chancellor
3. FA & CAO

4. Registrar
5. All Deans
6. All HODS

Notice Board: University of Engineering & Management, Kolkata
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ONGOING

GRANT-IN AID PROJECTS

DEPT OF CST & CSIT

- 1) EdgeSentinel: AI Enabled Patient Monitoring System with Edge Computing and Enhanced Security.

No. of Grant-In-Aid
Projects running under
IEM Trust : 28.

- 2) IoT Solutions for Paralyzed Patient Care
- 3) Automated Sound-Based Object Searching Hardware Module.
- 4) LANDEASE
- 5) Solar Guide Auto Parking
- 6) Low Light or Nighttime Object Detection from Fused Ultrasonic and IR/Radar Image Sensing utilizing Deep Learning.
- 7) EcoPath: A Platform for Environmentally Sustainable Road Planning.
- 8) Investigation of Compressor Oil-based Hybrid Nanolubricants for Enhanced Refrigeration Performance using Machine Learning.